

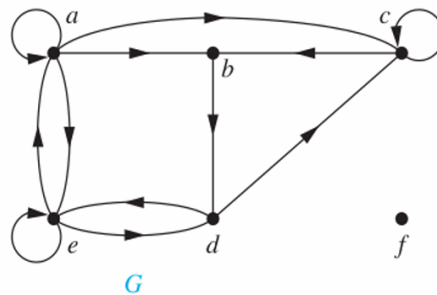
Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Quiz Exam

Full Marks: 10 (2 x 5)

[Answer any 2 questions]

Time: 20 mins

1. Encrypt the plaintext message “I LOVE MY COUNTRY” using the shift cipher with shift key $k = 17$.
2. Using the Fast Modular Exponentiation method, find $7^{222} \bmod 11$.
3. Find the in-degree and out-degree of each vertex in the graph G with directed edges.



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[Answer any 2 questions]

Time: 20 mins

1. Encrypt the plaintext message “STOP GLOBAL WARMING” using the shift cipher with shift key $k = 19$.
2. Using the Fast Modular Exponentiation method, find $3^{644} \bmod 645$.
3. Using Dijkstra’s algorithm, find the shortest path and the length between a and z of the following weighted graph.

